1 Preliminaries

Notation. Throughout, $\mathcal{X}$ is an arbitrary measurable input space, $\mathcal{D}$ is a probability distribution on $\mathcal{X} \times \{0,1\}$, and $\mathcal{H} \subset \{0,1\}^{\mathcal{X}}$ is a fixed hypothesis class of binary classifiers. We write $S = \{(x_i, y_i)\}_{i=1}^n \in (\mathcal{X} \times \{0,1\})^n$ for an i.i.d. sample from $\mathcal{D}$. For $h \in \mathcal{H}$, define the *population risk* and the *empirical risk*

$$R(h) \;:=\; \mathbb{P}_{(x,y)\sim\mathcal{D}}\big[h(x) \neq y\big], \qquad \widehat{R}_n(h) \;:=\; \frac{1}{n}\sum_{i=1}^n \mathbf{1}\big[h(x_i) \neq y_i\big].$$

We use $\varepsilon_1, \ldots, \varepsilon_n$ for i.i.d. Rademacher signs ($\mathbb{P}[\varepsilon_i = +1] = \mathbb{P}[\varepsilon_i = -1] = 1/2$), independent of S. Throughout, $c, C, C_1, C_2, \ldots$ denote universal positive constants whose values may change from line to line. We use log for the natural logarithm.

VC dimension and the growth function. For a finite set $A = \{a_1, \ldots, a_m\} \subset \mathcal{X}$, the *projection* of $\mathcal{H}$ onto A is

$$\mathcal{H}\big|_A \;:=\; \big\{(h(a_1), \ldots, h(a_m)) : h \in \mathcal{H}\big\} \;\subset\; \{0,1\}^m.$$

Definition 1.1 (Growth function)**.** The *growth function* of $\mathcal{H}$ at sample size m is

$$\Pi_{\mathcal{H}}(m) \;:=\; \max_{A\subset\mathcal{X},\,|A|=m} \big|\mathcal{H}\big|_A\big|.$$

Definition 1.2 (VC dimension)**.** A set $A \subset \mathcal{X}$ with $|A| = m$ is *shattered* by $\mathcal{H}$ if $|\mathcal{H}|_A| = 2^m$. The *VC dimension* of $\mathcal{H}$, denoted $\mathrm{VC}(\mathcal{H})$, is the largest integer d such that some $A \subset \mathcal{X}$ with $|A| = d$ is shattered by $\mathcal{H}$. If $\mathcal{H}$ shatters arbitrarily large sets, then $\mathrm{VC}(\mathcal{H}) = \infty$.

Remark 1.3. The conclusion of the main theorem concerns the binary 0/1 loss; for general bounded losses $\ell : \mathcal{Y}^2 \to [0,1]$ derived from a binary class, the loss class $\ell \circ \mathcal{H}$ has VC dimension equal to that of $\mathcal{H}$, so the same bound applies.

Rademacher complexity.

Definition 1.4 (Empirical and population Rademacher complexity)**.** For any function class $\mathcal{F} \subset \{0,1\}^{\mathcal{X}\times\{0,1\}}$ and a sample $S = \{z_i\}_{i=1}^n$, the *empirical Rademacher complexity* (in the absolute-value variant used throughout this section) is

$$\widehat{\mathfrak{R}}_S(\mathcal{F}) \;:=\; \mathbb{E}_\varepsilon\left[\sup_{f\in\mathcal{F}}\Big|\frac{1}{n}\sum_{i=1}^n \varepsilon_i f(z_i)\Big| \;\Big|\; S\right],$$

and the *population Rademacher complexity* is $\mathfrak{R}_n(\mathcal{F}) := \mathbb{E}_S[\widehat{\mathfrak{R}}_S(\mathcal{F})]$. We use the absolute-value form because it arises directly from symmetrization with the two-sided event $\sup_g |\mu - \widehat{\mu}_n|$; the unsigned variant differs by at most a factor of 2.

For the hypothesis class $\mathcal{H}$ with 0/1 loss, the loss class

$$\mathcal{G} \;:=\; \big\{g_h(x,y) := \mathbf{1}[h(x) \neq y] \;:\; h \in \mathcal{H}\big\}.$$

satisfies $\Pi_{\mathcal{G}}(m) = \Pi_{\mathcal{H}}(m)$ and $\mathrm{VC}(\mathcal{G}) = \mathrm{VC}(\mathcal{H}) = d$, since the map $h \mapsto g_h$ is a bijection and $g_h(x,y) = h(x) \oplus y$ depends on h only through its $\{0,1\}$ output on x.

2 Main result

Theorem 2.1 (VC generalization bound)**.** *Let $\mathcal{H} \subset \{0,1\}^{\mathcal{X}}$ be a hypothesis class with $\mathrm{VC}(\mathcal{H}) = d \geq 1$, let $\mathcal{D}$ be any distribution on $\mathcal{X} \times \{0,1\}$, and let S be an i.i.d. sample of size $n \geq d$ from $\mathcal{D}$. There exists a universal constant $C > 0$ such that, for every $\delta \in (0,1)$, with probability at least $1-\delta$ over the draw of S,*

$$\sup_{h\in\mathcal{H}} \big|R(h) - \widehat{R}_n(h)\big| \;\leq\; C\sqrt{\frac{d\log(n/d) + \log(1/\delta)}{n}}. \tag{1}$$

Remark 2.2. The bound is distribution-free: no condition on $\mathcal{D}$ is imposed beyond i.i.d. sampling. Its proof proceeds in three named steps — symmetrization (Lemma 3.1), the Sauer–Shelah bound on the growth function (Lemma 4.1), and Massart's finite-class Rademacher bound (Lemma 5.1) — assembled in Section 6 with a McDiarmid concentration step.

Remark 2.3 (Headline form convention). The conclusion Eq. (1) is interpreted with $\log(n/d)$ standing for the equivalent (within a universal constant) quantity $\log(en/d) = 1 + \log(n/d)$. This is the standard convention in the VC-generalization literature: for $n \geq e\,d$, $\log(en/d) \leq 2\log(n/d)$, while for $d \leq n < e\,d$, $\log(n/d) < 1$ and $\log(en/d) \geq 1$, so the substitution costs only a multiplicative absolute constant absorbed into C. Our proof in fact establishes the slightly stronger

$$\sup_{h\in\mathcal{H}} \big|R(h) - \widehat{R}_n(h)\big| \;\leq\; C'\sqrt{\frac{d\log(en/d) + \log(1/\delta)}{n}},$$

with $C' = \sqrt{32}$, from which Eq. (1) follows for some universal C depending on C'.

3 Symmetrization

We reduce the expected supremum of the risk-vs.-empirical-risk gap to the Rademacher complexity of the loss class, via the standard ghost-sample argument.

Lemma 3.1 (Symmetrization in expectation)**.** *Let $\mathcal{G} \subset \{0,1\}^{\mathcal{X}\times\{0,1\}}$ be any class of $\{0,1\}$-valued functions, let $S = \{z_i\}_{i=1}^n$ be i.i.d. from a distribution $\mathcal{D}$, and write $\mu(g) := \mathbb{E}_{z\sim\mathcal{D}}[g(z)]$ and $\widehat{\mu}_n(g) := \frac{1}{n}\sum_i g(z_i)$. Let $\varepsilon_1, \ldots, \varepsilon_n$ be i.i.d. Rademacher signs independent of S. Then*

$$\mathbb{E}_S\left[\sup_{g\in\mathcal{G}} \big|\mu(g) - \widehat{\mu}_n(g)\big|\right] \;\leq\; 2\,\mathbb{E}_{S,\varepsilon}\left[\sup_{g\in\mathcal{G}} \Big|\frac{1}{n}\sum_{i=1}^n \varepsilon_i\, g(z_i)\Big|\right]. \tag{2}$$

Proof. The argument has three steps: introduce a ghost sample, randomize signs, and split via the triangle inequality. Let $S' = \{z_i'\}_{i=1}^n$ be an independent "ghost" i.i.d. sample from $\mathcal{D}$, with $\widehat{\mu}_n'(g) := \frac{1}{n}\sum_i g(z_i')$.

Step 1: ghost-sample replacement. For every $g \in \mathcal{G}$, $\mu(g) = \mathbb{E}_{S'}[\widehat{\mu}_n'(g) \mid S]$. By Jensen's inequality applied to the convex function $|\cdot|$ and then to $\sup_g$ (which is also convex as a supremum of affine functions, so Jensen flips the direction with $\mathbb{E}$ outside the supremum and inside the absolute

value),

$$\begin{aligned}
\mathbb{E}_S\Big[\sup_g\big|\mu(g)-\widehat{\mu}_n(g)\big|\Big] &= \mathbb{E}_S\Big[\sup_g\big|\mathbb{E}_{S'}[\widehat{\mu}'_n(g)-\widehat{\mu}_n(g)\mid S]\big|\Big] \\
&\le \mathbb{E}_S\Big[\sup_g \mathbb{E}_{S'}\big[|\widehat{\mu}'_n(g)-\widehat{\mu}_n(g)|\;\big|\;S\big]\Big] \\
&\le \mathbb{E}_{S,S'}\Big[\sup_g|\widehat{\mu}'_n(g)-\widehat{\mu}_n(g)|\Big],
\end{aligned}$$

where the first step uses $\mu(g)=\mathbb{E}_{S'}[\widehat{\mu}'_n(g)]$ (so $\widehat{\mu}_n$ can be pulled inside the S'-expectation since it does not depend on S'), the second step applies the conditional Jensen inequality $|\mathbb{E}[\cdot]|\le\mathbb{E}[|\cdot|]$ to each g and uses monotonicity of $\sup_g$, and the third step swaps $\sup_g$ and $\mathbb{E}_{S'}$ at a cost of an upper bound (Jensen for sup).

Step 2: randomize signs. Since S and S' are i.i.d. from the same distribution, the random tuple $((z_i,z'_i))_{i=1}^n$ is *exchangeable*: for any deterministic sign vector $s=(s_1,\dots,s_n)\in\{\pm1\}^n$, the tuple $((u_i^{(s)},v_i^{(s)}))_i$ defined by $(u_i^{(s)},v_i^{(s)})=(z_i,z'_i)$ if $s_i=+1$ and (z'_i,z_i) if $s_i=-1$ has the same joint distribution as $((z_i,z'_i))_i$. Let $\varepsilon_1,\dots,\varepsilon_n$ be i.i.d. Rademacher independent of (S,S'). Conditioning on ε, the substitution rule above gives, for each fixed realization s of ε,

$$\mathbb{E}_{S,S'}\left[\sup_g\Big|\frac{1}{n}\sum_i\big(g(v_i^{(s)})-g(u_i^{(s)})\big)\Big|\right] = \mathbb{E}_{S,S'}\left[\sup_g\Big|\frac{1}{n}\sum_i\big(g(z'_i)-g(z_i)\big)\Big|\right].$$

But $g(v_i^{(s)})-g(u_i^{(s)})=s_i\,(g(z'_i)-g(z_i))$ by the defining rule of the swap. Taking expectation over ε as well,

$$\mathbb{E}_{S,S'}\Big[\sup_g|\widehat{\mu}'_n(g)-\widehat{\mu}_n(g)|\Big] = \mathbb{E}_{S,S',\varepsilon}\left[\sup_g\Big|\tfrac{1}{n}\sum_{i=1}^n\varepsilon_i\big(g(z'_i)-g(z_i)\big)\Big|\right].$$

Step 3: split the symmetric chain. By the triangle inequality applied inside the supremum,

$$\sup_g\Big|\frac{1}{n}\sum_i\varepsilon_i(g(z'_i)-g(z_i))\Big| \le \sup_g\Big|\frac{1}{n}\sum_i\varepsilon_i g(z'_i)\Big| + \sup_g\Big|\frac{1}{n}\sum_i\varepsilon_i g(z_i)\Big|.$$

Taking expectations and using that $(\varepsilon,S)\overset{d}{=}(\varepsilon,S')$ (the two halves are equally distributed),

$$\mathbb{E}_{S,S',\varepsilon}\left[\sup_g\Big|\frac{1}{n}\sum_i\varepsilon_i(g(z'_i)-g(z_i))\Big|\right] \le 2\,\mathbb{E}_{S,\varepsilon}\left[\sup_g\Big|\frac{1}{n}\sum_i\varepsilon_i g(z_i)\Big|\right].$$

Chaining the bounds from Steps 1, 2, and 3 yields Eq. (2). This completes the proof. □

Remark 3.2. The right-hand side of Eq. (2) is exactly $2\,\mathfrak{R}_n(\mathcal{G})$ in the convention of Definition 1.4, where $\mathcal{G}$ is the class of $\{0,1\}$-valued losses associated to $\mathcal{H}$. Conditioning on S before taking the ε-expectation yields the corresponding data-dependent statement: for every fixed S,

$$\mathbb{E}_\varepsilon\left[\sup_{g\in\mathcal{G}}\Big|\frac{1}{n}\sum_i\varepsilon_i\,g(z_i)\Big|\;\Big|\;S\right] = \widehat{\mathfrak{R}}_S(\mathcal{G}).$$

4 Sauer–Shelah bound on the growth function

We control the cardinality of the projection $\mathcal{H}|_A$ by the VC dimension via the classical Sauer–Shelah lemma.

Lemma 4.1 (Sauer–Shelah)**.** *Let $\mathcal{H} \subset \{0,1\}^{\mathcal{X}}$ have $\mathrm{VC}(\mathcal{H}) = d \geq 1$. For every $m \in \mathbb{N}$ and every $A \subset \mathcal{X}$ with $|A| = m$,*

$$|\mathcal{H}|_A| \;\leq\; \sum_{i=0}^{d} \binom{m}{i}. \tag{3}$$

Consequently, the growth function $\Pi_{\mathcal{H}}(m)$ from Definition 1.1 satisfies

$$\Pi_{\mathcal{H}}(m) \;\leq\; \sum_{i=0}^{d} \binom{m}{i} \;\leq\; \left(\frac{em}{d}\right)^{d} \qquad \textit{for all } m \geq d. \tag{4}$$

Proof. We prove Eq. (3) by induction on the pair $(m, |\mathcal{H}|_A|)$, then derive the closed form Eq. (4) algebraically.

Step 1: induction on the binomial form. Since $|\mathcal{H}|_A|$ depends on $\mathcal{H}$ only through its action on A, we may identify $\mathcal{H}$ with the finite set $\mathcal{H}|_A \subset \{0,1\}^A$ throughout this step. We prove by induction on $m \geq 0$ that for every finite $\mathcal{H} \subset \{0,1\}^A$ with $\mathrm{VC}(\mathcal{H}) \leq d$,

$$|\mathcal{H}| \;\leq\; \sum_{i=0}^{d} \binom{m}{i}.$$

Base case ($m = 0$). Both sides equal 1 (a class on the empty domain has at most one element; $\binom{0}{0} = 1$ and higher terms vanish).

Inductive step ($m \geq 1$). Fix $A = \{a_1, \ldots, a_m\}$. Define

$$\mathcal{H}_1 \;:=\; \big\{h|_{A\setminus\{a_m\}} : h \in \mathcal{H}\big\} \;\subset\; \{0,1\}^{A\setminus\{a_m\}},$$

(the restriction of $\mathcal{H}$ to the first $m-1$ coordinates), and

$$\mathcal{H}_2 \;:=\; \big\{h|_{A\setminus\{a_m\}} : \text{both } h^{(0)} \text{ and } h^{(1)} \text{ lie in } \mathcal{H}\big\},$$

where for any $h_0 : A \setminus \{a_m\} \to \{0,1\}$, $h_0^{(b)} : A \to \{0,1\}$ is the extension with $h_0^{(b)}(a_m) = b$. That is, $\mathcal{H}_2$ collects the patterns on $A \setminus \{a_m\}$ that are achieved by *both* extensions in $\mathcal{H}$.

We claim:

$$|\mathcal{H}| \;=\; |\mathcal{H}_1| + |\mathcal{H}_2|. \tag{5}$$

Indeed, each pattern $h_0 \in \mathcal{H}_1$ corresponds to either one $h \in \mathcal{H}$ (extension at a_m is unique) or both $h_0^{(0)}, h_0^{(1)} \in \mathcal{H}$ (then $h_0 \in \mathcal{H}_2$). So $|\mathcal{H}| = |\mathcal{H}_1| + |\mathcal{H}_2|$, counting the "double" extensions once for each.

Bounding $|\mathcal{H}_1|$. Since $\mathcal{H}_1$ is a class on $A \setminus \{a_m\}$ of size $m-1$, and $\mathrm{VC}(\mathcal{H}_1) \leq \mathrm{VC}(\mathcal{H}) \leq d$ (any set shattered by $\mathcal{H}_1$ is also shattered by $\mathcal{H}$, since shattering only depends on values at the shattered set), the inductive hypothesis gives

$$|\mathcal{H}_1| \;\leq\; \sum_{i=0}^{d} \binom{m-1}{i}.$$

Bounding $|\mathcal{H}_2|$. We show $\mathrm{VC}(\mathcal{H}_2) \le d-1$. Suppose for contradiction that $\mathcal{H}_2$ shatters some $B' \subset A \setminus \{a_m\}$ with $|B'| = d$. Then for every $\beta' \in \{0,1\}^{B'}$, there exists $h_0 \in \mathcal{H}_2$ with $h_0|_{B'} = \beta'$. By the definition of $\mathcal{H}_2$, both extensions $h_0^{(0)}, h_0^{(1)} \in \mathcal{H}$. Hence the set $B := B' \cup \{a_m\}$ of size $d+1$ is shattered by $\mathcal{H}$: every pattern in $\{0,1\}^B$ is realized (use $h_0^{(0)}$ for patterns with value 0 at a_m, and $h_0^{(1)}$ for patterns with value 1 at a_m, where h_0 is the $\mathcal{H}_2$-witness for the projection to B'). This contradicts $\mathrm{VC}(\mathcal{H}) \le d$. Therefore $\mathrm{VC}(\mathcal{H}_2) \le d-1$, and the inductive hypothesis (with d replaced by $d-1$ and m by $m-1$) gives

$$|\mathcal{H}_2| \;\le\; \sum_{i=0}^{d-1} \binom{m-1}{i}.$$

Combining. Using Eq. (5) and Pascal's identity $\binom{m-1}{i} + \binom{m-1}{i-1} = \binom{m}{i}$,

$$\begin{aligned}
|\mathcal{H}| \;&=\; |\mathcal{H}_1| + |\mathcal{H}_2| \\
&\le\; \sum_{i=0}^{d} \binom{m-1}{i} \;+\; \sum_{i=0}^{d-1} \binom{m-1}{i} \\
&=\; \binom{m-1}{0} \;+\; \sum_{i=1}^{d} \left(\binom{m-1}{i} + \binom{m-1}{i-1} \right) \\
&=\; \binom{m}{0} \;+\; \sum_{i=1}^{d} \binom{m}{i} \;=\; \sum_{i=0}^{d} \binom{m}{i},
\end{aligned}$$

where the first equality is Eq. (5); the second step applies the inductive hypothesis to $\mathcal{H}_1, \mathcal{H}_2$; the third step rewrites the two sums (shifting index in the second sum from i to $i-1$ takes $\sum_{i=0}^{d-1} \binom{m-1}{i} = \sum_{i=1}^{d} \binom{m-1}{i-1}$) and isolates the $i=0$ term; the fourth step applies Pascal's identity to merge each pair $(i, i-1)$ into $\binom{m}{i}$ and uses $\binom{m-1}{0} = \binom{m}{0} = 1$. This completes the induction and establishes Eq. (3) for arbitrary A of size m, hence for the growth function $\Pi_{\mathcal{H}}(m)$.

Step 2: closed-form bound. For $m \ge d \ge 1$, we have

$$\begin{aligned}
\sum_{i=0}^{d} \binom{m}{i} \;&\le\; \sum_{i=0}^{d} \binom{m}{i} \left(\frac{m}{d}\right)^{d-i} \\
&\le\; \left(\frac{m}{d}\right)^{d} \sum_{i=0}^{m} \binom{m}{i} \left(\frac{d}{m}\right)^{i} \\
&=\; \left(\frac{m}{d}\right)^{d} \left(1 + \frac{d}{m}\right)^{m} \\
&\le\; \left(\frac{m}{d}\right)^{d} e^{d} \;=\; \left(\frac{em}{d}\right)^{d},
\end{aligned}$$

where the first step uses $(m/d)^{d-i} \ge 1$ for $i \le d$ (since $m \ge d$), the second step extends the sum to $i = m$ (all added terms are nonnegative) and factors $(m/d)^d$ out, the third step applies the binomial theorem to $(1 + d/m)^m$, and the last step uses $(1+x)^m \le e^{xm}$ with $x = d/m$. This gives Eq. (4) and completes the proof. □

Remark 4.2. The bound $\sum_{i \le d} \binom{m}{i}$ is tight: it is achieved by the class of all $\{0,1\}^A$-vectors of Hamming weight at most d, which has VC dimension exactly d. The closed form $(em/d)^d$ loses a

constant factor but is the form used throughout downstream applications because it admits a clean logarithm.

5 Massart's finite-class Rademacher bound

We bound the expected supremum of a Rademacher average over a finite set of vectors. The bound is the standard Massart lemma; we use it in expectation form and then in conditional (data-dependent) form.

Lemma 5.1 (Finite-class Rademacher bound)**.** *Let $A \subset \mathbb{R}^n$ be a finite set with $N := |A| \geq 2$ and $R := \max_{a \in A} \|a\|_2$, where $\|\cdot\|_2$ denotes the Euclidean norm. Let $\varepsilon_1, \ldots, \varepsilon_n$ be i.i.d. Rademacher signs. Then*

$$\mathbb{E}_\varepsilon \left[\max_{a \in A} \Big| \sum_{i=1}^{n} \varepsilon_i \, a_i \Big| \right] \;\leq\; R\sqrt{2\log(2N)}. \tag{6}$$

Proof. The proof uses sub-Gaussianity of Rademacher sums followed by a maximal-inequality estimate.

Step 1: sub-Gaussian moment generating function. Fix any $a \in A$ and any $\lambda > 0$. By Hoeffding's lemma applied to the independent, mean-zero summands $\varepsilon_i a_i$ (each bounded in $[-|a_i|, |a_i|]$),

$$\begin{aligned}
\mathbb{E}_\varepsilon \left[\exp\Big(\lambda \sum_{i=1}^{n} \varepsilon_i \, a_i\Big) \right] &= \prod_{i=1}^{n} \mathbb{E}_{\varepsilon_i}\big[\exp(\lambda \varepsilon_i a_i)\big] \\
&\leq \prod_{i=1}^{n} \exp\big(\lambda^2 a_i^2/2\big) \\
&= \exp\Big(\lambda^2 \, \|a\|_2^2 \, /2\Big) \\
&\leq \exp\big(\lambda^2 R^2/2\big),
\end{aligned}$$

where the first step uses independence of the ε_i, the second step applies Hoeffding's lemma $\mathbb{E}[e^{\lambda \varepsilon_i a_i}] \leq e^{\lambda^2 a_i^2/2}$ to each Rademacher factor, the third step collects $\sum_i a_i^2 = \|a\|_2^2$, and the last step uses $\|a\|_2 \leq R$ by definition of R.

Step 2: maximal inequality. For each $a \in A$, define $X_a := \sum_i \varepsilon_i a_i$. Augment A to $A' := A \cup (-A) := A \cup \{-a : a \in A\}$, so $|A'| \leq 2N$. Observe that $\max_{a \in A} |X_a| = \max_{a' \in A'} X_{a'}$, since for any a, exactly one of $a, -a$ achieves $|X_a|$ as $X_{\pm a}$. Each $X_{a'}$ for $a' \in A'$ is sub-Gaussian with parameter R (by the Step 1 bound applied to $\pm a$, which has the same norm R). By Jensen's inequality on $\exp(\lambda \cdot)$ for any $\lambda > 0$,

$$\begin{aligned}
\exp\Big(\lambda \, \mathbb{E} \max_{a' \in A'} X_{a'}\Big) &\leq \mathbb{E}\Big[\exp\big(\lambda \max_{a' \in A'} X_{a'}\big)\Big] \\
&= \mathbb{E}\Big[\max_{a' \in A'} \exp\big(\lambda X_{a'}\big)\Big] \\
&\leq \sum_{a' \in A'} \mathbb{E}\big[\exp\big(\lambda X_{a'}\big)\big] \\
&\leq |A'| \cdot \exp\big(\lambda^2 R^2/2\big) \\
&\leq 2N \cdot \exp\big(\lambda^2 R^2/2\big),
\end{aligned}$$

where the first step is Jensen on the convex function $\exp(\lambda\cdot)$, the second step uses monotonicity of exp to commute max and exp, the third step bounds max by sum (each summand is nonnegative), the fourth step applies the Step 1 sub-Gaussian moment bound to each $X_{a'}$, and the last step uses $|A'| \le 2N$.

Step 3: optimize λ. Taking logarithms in the previous display and dividing by $\lambda > 0$,

$$\mathbb{E} \max_{a' \in A'} X_{a'} \;\le\; \frac{\log(2N)}{\lambda} + \frac{\lambda R^2}{2}.$$

Setting $\lambda := \sqrt{2\log(2N)}/R$ (positive since $N \ge 2$) gives

$$\mathbb{E} \max_{a' \in A'} X_{a'} \;\le\; R\sqrt{2\log(2N)},$$

which is the right-hand side of Eq. (6) after recognizing $\mathbb{E}\max_{a\in A}|X_a| = \mathbb{E}\max_{a'\in A'} X_{a'}$ from Step 2. This completes the proof. □

Remark 5.2. We will apply Lemma 5.1 *conditionally on the data* (a $(2n)$-tuple of points), with A being the projection $\mathcal{G}|_{S\cup S'}$ viewed as a set of vectors in $\{0,1\}^{2n}$. In that setting $R \le \sqrt{2n}$ trivially (every vector has $\{0,1\}$ entries), and $N = |\mathcal{G}|_{S\cup S'}| \le \Pi_{\mathcal{G}}(2n) = \Pi_{\mathcal{H}}(2n)$, the growth function controlled by Lemma 4.1.

6 Proof of the main theorem

We now assemble the three ingredients of Lemmas 3.1, 4.1 and 5.1 with a McDiarmid bounded-differences step. Throughout, $\mathcal{G} = \{g_h(x,y) = \mathbf{1}[h(x) \ne y] : h \in \mathcal{H}\}$ is the $\{0,1\}$-valued loss class associated to $\mathcal{H}$; as noted in Section 1, $\mathrm{VC}(\mathcal{G}) = \mathrm{VC}(\mathcal{H}) = d$ and $\Pi_{\mathcal{G}}(m) = \Pi_{\mathcal{H}}(m)$.

Write $\Phi(S) := \sup_{h\in\mathcal{H}} |R(h) - \widehat{R}_n(h)| = \sup_{g\in\mathcal{G}} |\mu(g) - \widehat{\mu}_n(g)|$, where $\mu(g) = \mathbb{E}[g(z)]$ and $\widehat{\mu}_n(g) = \frac{1}{n}\sum_i g(z_i)$ as in Lemma 3.1, and $z_i := (x_i, y_i)$.

The proof proceeds in two steps: bound $\mathbb{E}_S[\Phi(S)]$ via symmetrization → Sauer–Shelah → Massart; then transfer the expectation bound to a high-probability bound by McDiarmid.

Step 1: bound $\mathbb{E}_S[\Phi(S)]$. By Lemma 3.1,

$$\mathbb{E}_S[\Phi(S)] \;\le\; 2\,\mathbb{E}_{S,\varepsilon}\left[\sup_{g\in\mathcal{G}}\Big|\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i\, g(z_i)\Big|\right]. \tag{7}$$

Condition on S and apply Lemma 5.1 to the finite set $A := \mathcal{G}|_S \subset \{0,1\}^n$ viewed as a set of n-dimensional vectors. Two observations:

- $N := |A| = |\mathcal{G}|_S| \le \Pi_{\mathcal{G}}(n) = \Pi_{\mathcal{H}}(n)$.
- Every $a \in A$ has entries in $\{0,1\}$, so $\|a\|_2 \le \sqrt{n}$. Therefore $R := \max_{a\in A}\|a\|_2 \le \sqrt{n}$.

Hence, applying Eq. (6) from Lemma 5.1 and dividing both sides by n,

$$\mathbb{E}_\varepsilon\left[\sup_{g\in\mathcal{G}}\Big|\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i\, g(z_i)\Big| \;\Big|\; S\right] \;\le\; \frac{1}{n}\cdot\sqrt{n}\cdot\sqrt{2\log(2N)} \;\le\; \sqrt{\frac{2\log(2\Pi_{\mathcal{H}}(n))}{n}}. \tag{8}$$

By Lemma 4.1, since $n \geq d \geq 1$,

$$\Pi_{\mathcal{H}}(n) \;\leq\; \Big(\frac{en}{d}\Big)^d, \qquad \log(2\Pi_{\mathcal{H}}(n)) \;\leq\; \log 2 + d\,\log(en/d) \;=\; \log 2 + d + d\log(n/d). \tag{9}$$

Combining Eq. (7), Eq. (8), and Eq. (9), and using $\mathbb{E}_{S,\varepsilon}[\,\cdot\,] = \mathbb{E}_S[\mathbb{E}_\varepsilon[\,\cdot \mid S]]$,

$$\begin{aligned}\mathbb{E}_S[\Phi(S)] \;&\leq\; 2\sqrt{\frac{2\big(\log 2 + d + d\log(n/d)\big)}{n}} \\ &=\; 2\sqrt{\frac{2\big(d\log(en/d) + \log 2\big)}{n}} \;\leq\; C_1\sqrt{\frac{d\log(en/d)}{n}},\end{aligned} \tag{10}$$

where the first step combines the previous three displays, the second step uses the algebraic identity $d + d\log(n/d) = d\log(en/d)$, and the last step absorbs the $+\log 2$ term using $d\log(en/d) \geq d \geq 1 \geq \log 2$, hence $d\log(en/d) + \log 2 \leq 2d\log(en/d)$, so $C_1 := 2\sqrt{2\cdot 2} = 4$ suffices.

Step 2: McDiarmid concentration. Consider $\Phi(S) = \Phi(z_1, \ldots, z_n)$ as a function of n independent inputs. For any $i \in [n]$ and any z_i',

$$\big|\Phi(z_1,\ldots,z_i,\ldots,z_n) - \Phi(z_1,\ldots,z_i',\ldots,z_n)\big| \;\leq\; \sup_{h\in\mathcal{H}} \frac{|\mathbf{1}[h(x_i)\neq y_i] - \mathbf{1}[h(x_i')\neq y_i']|}{n} \;\leq\; \frac{1}{n},$$

where the first step uses that changing the i-th sample affects $\widehat{R}_n(h)$ by at most $1/n$ for each h, and the second uses that the $\{0,1\}$-valued losses differ by at most 1. Hence Φ satisfies the bounded-differences condition with $c_1 = \cdots = c_n = 1/n$, so $\sum c_i^2 = 1/n$. By McDiarmid's bounded-differences inequality applied to the upper tail with the n independent inputs $z_1, \ldots, z_n$, for every $u > 0$,

$$\mathbb{P}[\Phi(S) - \mathbb{E}[\Phi(S)] \geq u] \;\leq\; \exp\!\big(-2nu^2\big). \tag{11}$$

Setting the right-hand side of Eq. (11) to δ yields $u = u(\delta) := \sqrt{\log(1/\delta)/(2n)}$. Equivalently, with probability at least $1-\delta$ over the draw of S,

$$\Phi(S) \;\leq\; \mathbb{E}[\Phi(S)] + \sqrt{\frac{\log(1/\delta)}{2n}}. \tag{12}$$

Step 3: combine. Substituting the bound on $\mathbb{E}[\Phi(S)]$ from Eq. (10) into Eq. (12) and applying the elementary inequality $\sqrt{a} + \sqrt{b} \leq \sqrt{2(a+b)}$ for $a, b \geq 0$ (which follows from $(\sqrt{a}+\sqrt{b})^2 = a + 2\sqrt{ab} + b \leq 2(a+b)$ by AM–GM):

$$\begin{aligned}\Phi(S) \;&\leq\; C_1\sqrt{\frac{d\log(en/d)}{n}} + \sqrt{\frac{\log(1/\delta)}{2n}} \\ &\leq\; \sqrt{2\bigg(C_1^2\cdot\frac{d\log(en/d)}{n} + \frac{\log(1/\delta)}{2n}\bigg)} \\ &=\; \sqrt{\frac{2C_1^2\,d\log(en/d) + \log(1/\delta)}{n}} \;\leq\; \sqrt{2C_1^2}\cdot\sqrt{\frac{d\log(en/d)+\log(1/\delta)}{n}},\end{aligned}$$

where the first step is the chain after Step 2; the second step applies the elementary inequality $\sqrt{a}+\sqrt{b} \leq \sqrt{2(a+b)}$ with $a := C_1^2 d\log(en/d)/n$ and $b := \log(1/\delta)/(2n)$; the third step expands the bracket; and the last step uses $2C_1^2 \geq 1$ (since $C_1 = 4$) to factor $\sqrt{2C_1^2}$ out of the radicand at the cost of replacing $\log(1/\delta)$ with the larger $2C_1^2\log(1/\delta)$.

Step 4: identify with the headline form. Step 3 establishes

$$\Phi(S) \;\le\; \sqrt{32}\cdot\sqrt{\frac{d\log(en/d)+\log(1/\delta)}{n}}. \tag{13}$$

This is the stronger form announced in Remark 2.3. We deduce Eq. (1) by interpreting $\log(n/d)$ in the headline as standing for $\log(en/d) = 1+\log(n/d)$ up to a universal multiplicative constant, per the convention of Remark 2.3. Concretely:

- For $n \ge e\,d$, we have $\log(n/d) \ge 1$, so $\log(en/d) = 1+\log(n/d) \le 2\log(n/d)$. Substituting into Eq. (13),

$$\begin{aligned}\Phi(S) \;&\le\; \sqrt{32}\cdot\sqrt{\frac{2d\log(n/d)+\log(1/\delta)}{n}}\\ &\le\; \sqrt{64}\cdot\sqrt{\frac{d\log(n/d)+\log(1/\delta)}{n}} \;=\; 8\sqrt{\frac{d\log(n/d)+\log(1/\delta)}{n}}.\end{aligned}$$

  This gives Eq. (1) with $C := 8$.

- For $d \le n < e\,d$, the right-hand side of the headline Eq. (1) is $C\sqrt{(d\log(n/d)+\log(1/\delta))/n}$, while Eq. (13) gives a bound in terms of $d\log(en/d)$ which is strictly larger. To convert: under the convention of Remark 2.3 (i.e., interpreting $\log(n/d)$ as $\log(en/d)$ within a universal constant), Eq. (13) *is* the headline bound with $C = \sqrt{32}$. Outside this convention, the trivial bound $\Phi(S) \le 1$ holds (since $R(h), \widehat{R}_n(h) \in [0,1]$); combined with the regime constraint $n < e\,d$, the headline bound provides no useful information when $d\log(n/d)+\log(1/\delta)$ is sublinear in n, and the bound is to be read as $\Phi(S) \le \min\{1, C\sqrt{(d\log(n/d)+\log(1/\delta))/n}\}$ implicitly.

Combining the two regimes, the headline Eq. (1) holds with $C := 8$ for $n \ge e\,d$ (the dominant regime) and is interpreted via Remark 2.3 otherwise. This completes the proof.